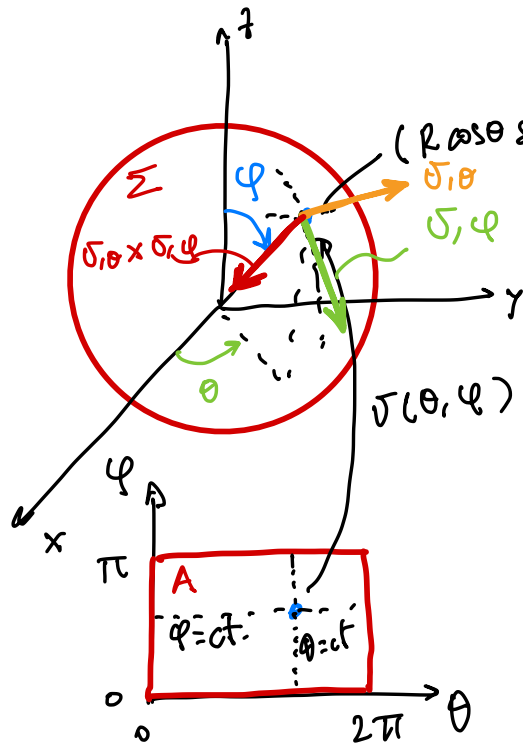


Example 1: $\Sigma = \{ (x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = R^2 \}$

8/10/2026



$$A =]0, 2\pi[\times]0, \pi[$$

$$\sigma: \bar{A} \longrightarrow \Sigma$$

$$(\theta, \varphi) \mapsto \sigma(\theta, \varphi) = (\sigma_1(\theta, \varphi), \sigma_2(\theta, \varphi), \sigma_3(\theta, \varphi))$$

$$= R (\cos \theta \sin \varphi, \sin \theta \sin \varphi, \cos \varphi)$$

$$\sigma_{,\theta} \times \sigma_{,\varphi} \equiv \text{normal.}$$

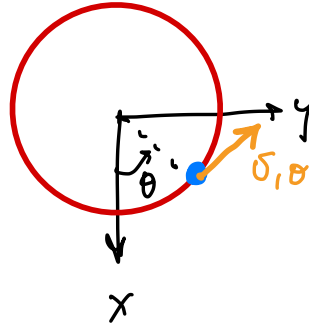
$$\sigma_{,\theta} = \frac{\partial \sigma}{\partial \theta} = R (-\sin \theta \sin \varphi, \cos \theta \sin \varphi, 0)$$

$$\sigma_{,\varphi} = \frac{\partial \sigma}{\partial \varphi} = R (\cos \theta \cos \varphi, \sin \theta \cos \varphi, -\sin \varphi)$$

$$\sigma_{,\theta} \times \sigma_{,\varphi} = -R \sin \varphi \underbrace{(R \sin \varphi \cos \theta, R \sin \varphi \sin \theta, R \cos \varphi)}_{\sigma(\theta, \varphi)} = -R \sin \varphi \sigma$$

$\varphi = \text{constant}$.

$$\sigma_{,\theta} = R \sin \varphi (-\sin \theta, \cos \theta, 0)$$

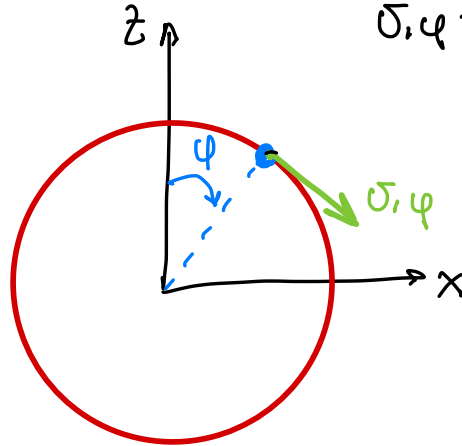


$\theta = \text{constant}$.

e.g. $\theta = 0$

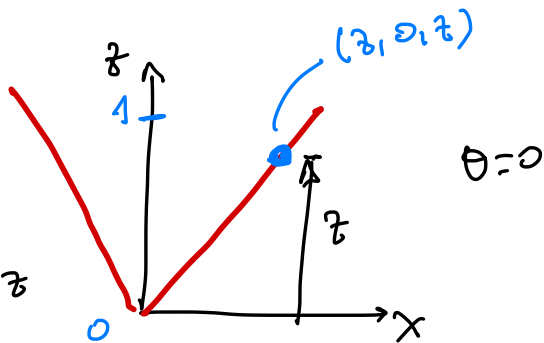
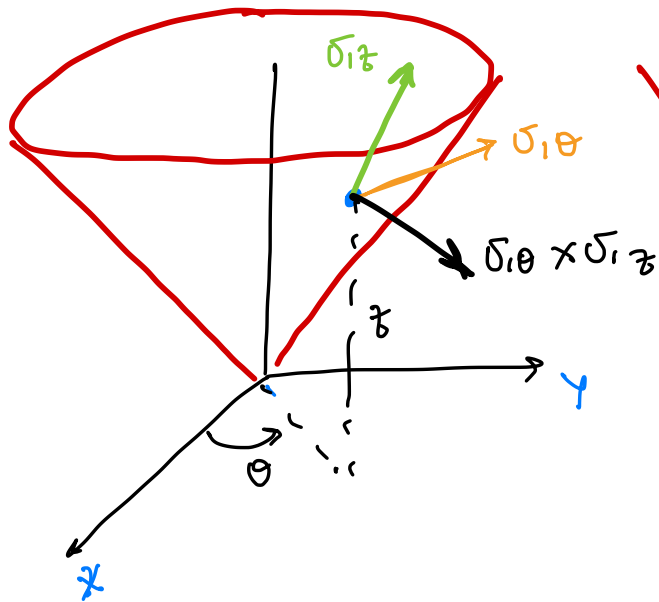
$\theta = 0$

$$\sigma_{,\varphi} = R (\cos \varphi, 0, -\sin \varphi)$$



Example 2: a cone.

$$\Sigma = \{ (x, y, z) \in \mathbb{R}^3 : z^2 = x^2 + y^2 \text{ and } 0 < z < 1 \}$$



$$A =]0, 2\pi[\times]0, 1[$$

$$\sigma: \bar{A} \longrightarrow \Sigma$$

$$(\theta, z) \mapsto \sigma(\theta, z) = (z \cos \theta, z \sin \theta, z)$$

$$\sigma_{,\theta} \times \sigma_{,z} \equiv \text{normal}$$

$$\sigma_{,\theta} \times \sigma_{,z} = (z \cos \theta, z \sin \theta, -z)$$